Deductive reasoning

1

a An argument is valid when its conclusion is guaranteed by the given set of premises. In other words, there is no case in which the premises are true but the conclusion is false.

b An argument is invalid when its conclusion is not necessarily guaranteed by the given set of premises. In other words, there is at least one case where the premises are true but the conclusion is false.

2 An invalid argument

3

a Fallacy of the Inverse

b Law of Detachment

c Disjunctive Syllogism

d Fallacy of the Converse

e Law of Syllogism

f Law of Contraposition

4

a Invalid

b Valid

c Valid

d Valid

e Valid

f Valid

g Invalid

5

a

i
$$\begin{array}{l} e \implies f \\ \sim e \\ \hline \therefore \ \sim f \end{array}$$

ii Invalid

b

i
$$\begin{array}{l} l \implies m \\ m \\ \hline \therefore \ l \end{array}$$

ii Invalid

c

i
$$\begin{array}{l} j \implies k \\ j \\ \hline \therefore \ k \end{array}$$

ii Valid

d

i
$$\begin{array}{l} a \implies b \\ b \implies c \\ \hline \therefore \ a \implies c \end{array}$$

ii Valid

e

i
$$\begin{array}{l} a \implies b \\ \sim b \\ \hline \therefore \ \sim a \end{array}$$

ii Valid

f

i
$$\begin{array}{l} c \implies d \\ c \\ \hline \therefore \ d \end{array}$$

ii Invalid

g

i
$$\begin{array}{r} j \implies k \\ j \\ \hline \therefore\ k \end{array}$$

ii Valid

h

i
$$\begin{array}{r} r \implies s \\ \sim s \\ \hline \therefore\ \sim r \end{array}$$

ii Valid

i

i
$$\begin{array}{r} a \implies b \\ b \implies c \\ \hline \therefore\ a \implies c \end{array}$$

ii Valid

j

i
$$\begin{array}{r} a \implies b \\ \sim a \\ \hline \therefore\ \sim b \end{array}$$

ii Invalid

6

a

i
$$\begin{array}{r} a \implies b \\ \sim b \\ \hline \therefore\ \sim a \end{array}$$

ii Valid

b

i
$$\begin{array}{l} l \vee \sim m \\ b \\ \hline \therefore\ a \end{array}$$

ii Valid

c

i
$$\begin{array}{l} r \vee s \\ b \sim r \\ \hline \therefore\ s \end{array}$$

ii Valid

d

i
$$\begin{array}{l} g \implies h \\ \sim g \\ \hline \therefore\ \sim h \end{array}$$

ii Invalid

e

i
$$\begin{array}{l} t \implies u \\ u \\ \hline \therefore\ t \end{array}$$

ii Invalid

7

a Valid

b Valid

c Valid

d Not valid

e Not valid

8

a We cannot deduce what kind of animal Holly is.

b If $\overleftrightarrow{AB}$ and $\overleftrightarrow{XY}$ are a pair of lines in the plane, then they are intersecting.

9

a When a plippi forces a blubbit to leave its nest, a nugnug always follows.

b Luigi works from blueprints.

10 All snoozles are jibboos.
All lobies are jibboos.

11 Linus has four feet.
If a yahyok has four feet, then it is a plippi.
Linus breathes underwater.

Additional questions

12 If p, then q.

13

a All multiples of four are even numbers.

b T is a scalene triangle.

14

a If a regular polygon has five sides, then it has an interior angle sum of 540°.

b If the sum of the digits of a number is a multiple of 9, then the number is divisible by 3.

15

a Valid (by the Law of Detachment)

b Invalid

c Valid (by the Law of Syllogism)

16

a No. Since the statement "Ishaq plays a sport" is not the hypothesis of the other premise, we cannot make any conclusion.

b Yes. We can deduce that Cyrille will wake up early on New Years Day.

c Yes. We can deduce that Airon is attending a geography class.

17

a Valid (by the Law of Syllogism)

b Invalid

c Valid (by the Law of Detachment)

18 Uxia's argument cannot be valid because "has six legs" is not a conclusion of any of the given conditional statements. As such, the hypothesis, "Is a ladybug" cannot reach that conclusion with the given statements.

19 We can match the hypothesis "Q has diagonals which bisect each other" with the first premise to deduce that Q is a parallelogram. We can then match the hypothesis "Q has at least one right angle" with the second premise to deduce that the parallelogram Q is also a rectangle. Lastly, we can match the hypothesis "Q has four sides of equal length" with the third premise to deduce that the rectangle Q is also a square, which is the conclusion of the argument.

20

a This deduction could be invalid because it is possible that Violette will not require the whole hour to get to the park.

b The family's deduction will be valid if they also know that Violette will reach the park at noon.

21 For example:

- Puerto brings a basketball to school.
- Everyone who owns a basketball plays basketball.
- Puerto only brings his own belongings to school.

22

a This argument is valid because the conclusion is guaranteed by the given set of premises.

b The argument is not true because not all of the premises are true.

23 The Law of Syllogism connects two conditional statements together, while the Law of Detachment uses a hypothesis to draw a conclusion. This means that the Law of Detachment is required to make a conclusion from a set of premises.